Spectral Radius Minimization for Signed Squares of Cycles

[AUTHOR NAME]

[AFFILIATION]
[DEPARTMENT], [INSTITUTION]
[CITY], [COUNTRY]
Corresponding author: [CORRESPONDING AUTHOR]
Email: [EMAIL] ORCID: [ORCID]

Abstract

For even $n \geq 8$, let $C_n^2 = C_n(1,2)$ be the square of the cycle, with its distance-one and distance-two edges independently assigned signs, with no periodicity assumption on the signing, and let m_n be the minimum signed adjacency spectral radius. A distinguished twisted signing attains $\rho_-(n)$, where $\rho_-(n)^2 = 4 + 2\cos(2\pi/n) + 2\cos(4\pi/n)$, at every even order. We prove that equality holds exactly for $n \in \{8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 36, 38, 42, 44, 46\}$; equivalently, strict failure occurs precisely at 32, at 40, and at every even $n \geq 48$. Structurally, a period-eight reference phase has squared spectral edge $\eta < 8$, while its six-gap phase slip is the unique least-edge abnormal positive single gap and has an isolated rank-two squared edge $c_6 \in (\eta, 8)$. Charge-compatible separated copies of this slip, together with a discrete IMS estimate, prove eventual failure, while exact finite verification shows that continuous failure begins precisely at order 48. The finite portions are reduced to explicitly finite objects and independently verified using exact arithmetic, so floating-point evidence makes no spectral decision.

Keywords: signed graph; spectral radius; cycle square; switching equivalence; phase slip; computer-assisted proof

MSC 2020: 05C50 (primary); 05C22, 68V05 (secondary)

1 Introduction

Let $n \geq 8$ be even. We write

$$G_n = C_n^2 = C_n(1,2)$$

for the graph on $\mathbb{Z}/n\mathbb{Z}$ in which two vertices are adjacent when their cyclic distance is one or two. An edge signing is a map $\sigma : E(G_n) \to \{\pm 1\}$, and A_σ is the real symmetric matrix whose i, j entry is $\sigma(ij)$ on an edge and zero otherwise. Our problem is to determine

$$m_n = \min_{\sigma: E(G_n) \to \{\pm 1\}} \rho(A_\sigma).$$

Thus the already formed cycle square is being signed: the distance-one and distance-two edges receive signs independently. It is not the graph obtained by first signing a cycle and then squaring that signed object. Switching at vertices conjugates A_σ by a diagonal $\{\pm 1\}$-matrix, so the minimization naturally takes place over switching classes.

The problem belongs to the general signing program in spectral graph theory. Signings encode the new eigenvalues of two-lifts in the work of Bilu and Linial [3], while Belardo, Cioabă, Koolen, and

Wang ask explicitly which signatures minimize the adjacency spectral radius of a fixed connected graph [2]. Interlacing families give strong existence results for one side of the signed spectrum; in the bipartite setting spectral symmetry turns this into the two-sided control used to construct Ramanujan lifts [9]. The graphs C_n^2 are nonbipartite, however, so a one-sided eigenvalue estimate does not by itself control $\rho(A_\sigma)$. Here the graph is fixed at each order and the question is the exact two-sided optimum, rather than an existence bound uniform over a broad graph class.

The problem-specific starting point is Suvagiya's preprint [10, Conjecture 3]. For $C_n(1,2)$, it identifies four distinguished switching classes in which every quadrilateral is unbalanced, computes the spectra of the two twisted classes, verifies global optimality for even orders through 18, and formulates the all-even-order assertion as Conjecture 3. We determine exactly when that twisted candidate is globally minimizing, thereby resolving and disproving the conjecture. In particular, the problem, the flux coordinates, and the twisted spectral formula are inputs from that work, not novelty claims of the present paper.

The answer is unexpectedly nonmonotone. The proposed equality holds for every even order from 8 through 30, fails first at 32, returns at $34, 36, 38$, fails again at 40, and returns once more at $42, 44, 46$. Only from 48 onward does failure persist at every even order. The first counterexample therefore does not mark the beginning of continuous failure, and the two isolated failures cannot be inferred from the eventual phase-slip mechanism developed below.

To state the classification, set

$$\rho_-(n)^2 = 4 + 2\cos\frac{2\pi}{n} + 2\cos\frac{4\pi}{n}, \qquad \theta_n = \rho_-(n)^2.$$

The distinguished class has alternating triangle flux and negative Hamilton-cycle holonomy; a direct antiperiodic Fourier calculation shows that it attains $\rho_-(n)$ for every even $n \geq 8$. This gives the upper bound $m_n \leq \rho_-(n)$ at all orders. At an order where equality holds, the reverse inequality for every signing is a separate conclusion of exact exhaustion.

Theorem 1.1 (Complete classification)**.** *For every even $n \geq 8$, the distinguished twisted signing attains $\rho_-(n)$, and*

$$m_n < \rho_-(n) \quad \Longleftrightarrow \quad n = 32, \quad n = 40, \quad \textit{or} \quad n \geq 48.$$

Equivalently,

$$m_n = \rho_-(n) \quad \Longleftrightarrow \quad n \in \{8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 36, 38, 42, 44, 46\}.$$

Theorem 1.1 classifies the truth of the proposed equality. It neither classifies all minimizing signings at the valid orders nor determines the exact value of m_n at the failing orders.

Its finite component determines the irregular small-order truth table. A different part of the argument explains why failure is eventually uninterrupted: a low-spectral periodic phase supports localized elementary interfaces, and well-separated copies of those interfaces can be closed on large rings. The assertion that continuous failure begins at 48 uses both this analytic tail and an exact finite bridge; the interface theory alone gives no such sharp onset.

The structural comparison phase is period eight and is distinct from the twisted signing that attains $\rho_-(n)$. In its quadrilateral-flux word, the positive sites occur at spacing four. Replacing one spacing by a positive gap g creates a single interface; we call $q = g - 4$ its excess charge. For a cyclic collection of gaps, the global closing law is $\sum_j q_j \equiv n \pmod 8$, whereas the local translation sector carried by one charge is $\sigma_{\mathrm{sec}}(q) = q \pmod 4$. These congruences have different roles and are not interchangeable. In particular, one, two, and three copies of the gap-six interface, whose charge is two, supply the three nonzero even residue classes needed in the finite-ring constructions.

The reference squared spectral edge is

$$\eta = 4 + \sqrt{10 + 2\sqrt{5}}.$$

For the bilateral positive single-gap operator A_g, put $H_g = A_g^2$ and $E(g) = \sup \operatorname{Spec}(H_g)$. Define c_6 to be the unique root in the rational interval

$$\frac{7905369311620327}{10^{15}} < c_6 < \frac{7905369311620328}{10^{15}}$$

of the degree-ten polynomial

$$\begin{aligned} p_6(y) = {} & 16y^{10} - 520y^9 + 6913y^8 - 48448y^7 + 191768y^6 \\ & - 423904y^5 + 484528y^4 - 270464y^3 + 137856y^2 \\ & - 19968y + 256. \end{aligned}$$

The exact isolation is important: all later strict comparisons are made against rational endpoints, not against a decimal approximation.

Theorem 1.2 (Reference and single-gap spectral hierarchy)**.** *For both lifts and both orientations,*

$$E(4) = \eta < c_6 = E(6), \qquad \operatorname{Spec}_{\mathrm{ess}}(H_6) = \operatorname{Spec}(H_{\mathrm{ref}}), \qquad \dim \ker(H_6 - c_6) = 2.$$

Moreover, for every positive integer $g \notin \{4, 6\}$,

$$E(g) > c_6 + \frac{1}{250}.$$

The scope of Theorem 1.2 is precise: G6 is the unique minimizer among abnormal positive single gaps. It is not asserted to minimize over arbitrary multi-gap or finite-core interfaces. The strict single-gap hierarchy makes G6 the elementary low-cost defect above the period-eight bulk. Its charge allows one, two, or three separated copies to close in the required residue classes. Exact patch identification transfers the bilateral reference/G6 bounds to those finite rings, and a discrete IMS partition controls the squared spectral top between the patches. This proves failure for every even $n \geq 240$. Exact finite verification on $48 \leq n < 240$ then shows that continuous failure begins precisely at 48.

The computer-assisted part is organized as a proof by finite reduction, not as numerical evidence. Switching quotients and local interlacing reduce the relevant cases to finite exact objects. For $8 \leq n \leq 32$, complete switching-class decisions establish the lower bounds through 30 and an exact witness at 32. For $34, 36, 38, 42, 44, 46$, a parity-lifted finite state graph is proved sound, complete, and terminating, and every terminal case is closed exactly; a separate rational positive-definiteness certificate gives the witness at 40. Finally, rational positive-definiteness certificates cover every even order from 48 through 238. Independent checkers reconstruct the finite objects and repeat the integer, rational, or algebraic decisions. Exhaustive generation and local finite-state reductions have established precedents in graph theory [7, 5], while all-order classifications often combine an analytic tail with finite closure [8, 6].

Exact signed spectral-radius classifications are known in settings where the underlying graph is also allowed to vary, including unbalanced unicyclic and bicyclic graphs [1] and connected unbalanced signed graphs of fixed order [4]. Those results make it essential to keep the present novelty claim narrow. To the best of our knowledge, no previous work gives an all-order classification of the minimizing behavior over all real edge signings of the fixed circulant family $C_n(1, 2)$. The closest direct source remains Suvagiya's conjecture; the complete truth set and the elementary single-gap mechanism are the two contributions supplied here. The paper is arranged as follows. Section 2 sets up switching coordinates, the twisted attaining candidate, and the period-eight reference phase. Section 3 develops gaps, charges, and the two closing laws. Sections 4 and 5 prove the G6 edge theorem and the complete single-gap hierarchy. Section 6 passes from bilateral interfaces to finite rings by patch identification and discrete IMS localization. Section 7 closes the remaining orders by exact finite arguments, and Section 8 records the resulting perspective. The appendices collect the algebraic G6 certificate and the exact finite-classification boundary.

2 Switching Coordinates and the Reference Phase

2.1 Edge signings, switching, and Hamilton gauge

Write the vertices of $G_n = C_n(1,2)$ as $\mathbb{Z}/n\mathbb{Z}$. Let a_i be the sign of the distance-one edge $\{i, i+1\}$, and let b_i be the sign of the distance-two edge $\{i, i+2\}$, with all subscripts read cyclically. Switching by $\varepsilon = (\varepsilon_i)_{i\in\mathbb{Z}/n\mathbb{Z}} \in \{\pm1\}^n$ changes these signs to

$$a'_i = \varepsilon_i a_i \varepsilon_{i+1}, \qquad b'_i = \varepsilon_i b_i \varepsilon_{i+2}.$$

At the matrix level this is the orthogonal conjugacy $A_{\sigma'} = D_\varepsilon A_\sigma D_\varepsilon$, where $D_\varepsilon = \operatorname{diag}(\varepsilon_i)$. Switching therefore preserves the complete spectrum, not only the spectral radius.

Lemma 2.1 (Hamilton gauge)**.** *Every signing of G_n is switching equivalent to one for which*

$$a_0 = \cdots = a_{n-2} = 1, \qquad a_{n-1} = \alpha,$$

where

$$\alpha = \prod_{i=0}^{n-1} a_i \in \{\pm1\}.$$

The residual sign α, the Hamilton-cycle holonomy, is invariant under switching.

Proof. Set $\varepsilon_0 = 1$ and recursively choose $\varepsilon_{i+1} = \varepsilon_i a_i$ for $0 \le i \le n-2$. Then $a'_i = 1$ along the spanning path $0, 1, \ldots, n-1$. On the closing edge,

$$a'_{n-1} = \varepsilon_{n-1} a_{n-1} \varepsilon_0 = \prod_{i=0}^{n-1} a_i.$$

The product of the signs around a cycle is unchanged by switching because every vertex factor occurs twice. □

2.2 Flux words, holonomy, and the attaining candidate

Before fixing a gauge, define the triangle flux at i by

$$\tau_i = a_i a_{i+1} b_i.$$

Each switching factor again occurs twice, so τ_i is invariant. In the Hamilton gauge of Lemma 2.1, the actual distance-two signs are

$$b_i = \tau_i \quad (0 \le i \le n-3), \qquad b_{n-2} = \alpha\tau_{n-2}, \qquad b_{n-1} = \alpha\tau_{n-1}.$$

Consequently (τ, α) parametrizes the switching classes bijectively once the vertex labels and seam are fixed. It is often cleaner to absorb the closing sign into the boundary condition. More precisely, let

$$\mathcal{U}_n^\alpha = \{u : \mathbb{Z} \to \mathbb{C} : u_{j+n} = \alpha u_j\}.$$

On this n-dimensional space the signed adjacency takes the form

$$(A_\tau u)_i = u_{i-1} + u_{i+1} + \tau_{i-2} u_{i-2} + \tau_i u_{i+2}. \tag{1}$$

This twisted-boundary realization includes every edge that crosses the Hamilton-gauge seam and is unitarily identical to the original finite signed matrix. We suppress α from A_τ when the boundary space $\mathcal{U}_n^\alpha$ is displayed explicitly.

The quadrilateral-flux word is

$$Q_i = \tau_i \tau_{i+1}.$$

It satisfies $\prod_i Q_i = 1$. Conversely, a cyclic word $Q \in \{\pm 1\}^n$ has a cyclic τ-lift precisely when $\prod_i Q_i = 1$; after choosing τ_0, the recurrence

$$\tau_{i+1} = Q_i \tau_i$$

determines that lift. The two choices of τ_0 give τ and $-\tau$, while α remains independent. Thus the exact cyclic data are (τ, α), or equivalently (Q, τ_0, α). The reduced pair (Q, α) represents two switching classes, a distinction that disappears only after passing to the squared spectrum or spectral radius. The lift condition $\prod_i Q_i = 1$ and the Hamilton twist α are independent. If n is even and $(Du)_i = (-1)^i u_i$, then

$$A_{-\tau} = -D A_\tau D, \qquad H_{-\tau} = D H_\tau D, \qquad H_\tau := A_\tau^2. \tag{2}$$

The unsquared spectra of the two lifts are reflected through zero; their squared spectra and spectral radii agree.

Proposition 2.2 (Attainment by the twisted candidate)**.** *For every even $n \geq 8$, the class*

$$Q_i = -1 \quad (0 \leq i < n), \qquad \alpha = -1,$$

has a representative $\tau_i = (-1)^i$ satisfying

$$\rho(A_\tau)^2 = 4\cos^2 \frac{\pi}{n} + 4\cos^2 \frac{2\pi}{n} = 4 + 2\cos \frac{2\pi}{n} + 2\cos \frac{4\pi}{n} = \theta_n.$$

Consequently $m_n \leq \rho_-(n)$ for every even $n \geq 8$.

Proof. Because $\alpha = -1$, the allowed Fourier parameters on $\mathcal{U}_n^{-1}$ are

$$\vartheta_k = \frac{(2k+1)\pi}{n}, \qquad 0 \leq k < n.$$

Multiplication by $(-1)^j$ carries the mode $e^{ij\vartheta}$ to $e^{ij(\vartheta+\pi)}$. Hence the span of these two modes is invariant, and in the displayed order the restriction of A_τ is

$$B(\vartheta) = 2 \begin{pmatrix} \cos\vartheta & \cos 2\vartheta \\ \cos 2\vartheta & -\cos\vartheta \end{pmatrix}.$$

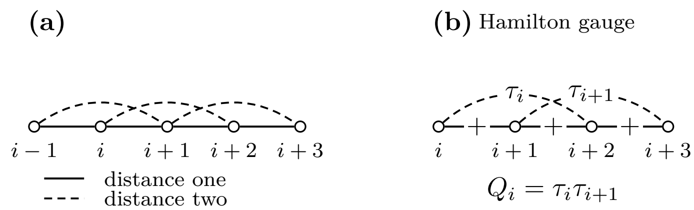

Figure 1: Switching coordinates for the cycle square. Distance-one and distance-two edges are signed independently; after Hamilton gauge, the local triangle word τ and its adjacent products $Q_i = \tau_i\tau_{i+1}$ record the flux data.

Thus

$$B(\vartheta)^2 = 4(\cos^2\vartheta + \cos^2 2\vartheta)I_2.$$

With $x = \cos 2\vartheta$, the quantity in parentheses is

$$F(x) = x^2 + \frac{x}{2} + \frac{1}{2}.$$

The relevant values lie in $[-1, \cos(2\pi/n)]$. Since F is convex, its maximum is at an endpoint. Now $F(-1) = 1$, whereas $n \geq 8$ gives

$$F\left(\cos\frac{2\pi}{n}\right) \geq F\left(\frac{1}{\sqrt{2}}\right) = 1 + \frac{1}{2\sqrt{2}} > 1.$$

The maximum therefore occurs at $\vartheta = \pi/n$ or $\pi - \pi/n$, which yields the first equality. The double-angle identity yields the second. Since this explicit signing is in the minimization domain, the final inequality follows. $\square$

The candidate in Proposition 2.2 is the signing that attains the conjectured comparison value. It is not the period-eight reference phase introduced in Subsection 2.4. The latter is a different low-spectral bulk used to construct phase slips.

2.3 The squared local operator

For a bilateral word τ, equation (1) defines a bounded self-adjoint range-two operator on $\ell^2(\mathbb{Z})$. Its square $H_\tau = A_\tau^2$ is the nonnegative range-four operator

$$\begin{aligned}(H_\tau u)_i = 4u_i &+ u_{i-2} + u_{i+2} \\ &+ \tau_{i-4}\tau_{i-2}u_{i-4} + (\tau_{i-3} + \tau_{i-2})u_{i-3} \\ &+ (\tau_{i-2} + \tau_{i-1})u_{i-1} + (\tau_{i-1} + \tau_i)u_{i+1} \\ &+ (\tau_i + \tau_{i+1})u_{i+3} + \tau_i\tau_{i+2}u_{i+4}. \end{aligned} \tag{3}$$

Indeed, this follows by inserting equation (1) four times into $A_\tau(A_\tau u)$ and collecting equal displacements. The extreme coefficients may also be written

$$\tau_{i-4}\tau_{i-2} = Q_{i-4}Q_{i-3}, \qquad \tau_i\tau_{i+2} = Q_iQ_{i+1},$$

while, for example, $\tau_i + \tau_{i+1} = \tau_i(1 + Q_i)$. Thus a negative Q_i cancels the associated odd-displacement coupling. Formula (3) is local and precedes every periodic or interface specialization.

Translation of the coefficient word is implemented by the bilateral shift. Reflection reverses the word and is unitary after the corresponding index change. Equation (2) handles the two lifts. We will use only these explicit equivalences in the body; the fiber-level zone-folding identities are recorded with the Floquet algebra in Appendix A.

2.4 The period-eight reference phase

The canonical reference word and its quadrilateral flux are

$$\tau_{\rm ref} = (+,+,-,+,-,-,+,-), \qquad Q_{\rm ref} = (+,-,-,-,+,-,-,-),$$

extended periodically. The positive $Q_{\rm ref}$-sites are exactly $4\mathbb{Z}$. More generally, for $s \in \mathbb{Z}/4\mathbb{Z}$, let B_s denote the translate whose positive Q-sites are $s + 4\mathbb{Z}$. Each B_s has two τ-lifts; since one four-site cell has Q-product -1, those lifts are antiperiodic over four sites and periodic over eight.

For a nonzero Bloch multiplier z, impose

$$u_{8m+r} = z^m v_r, \qquad 0 \le r < 8,$$

and write $A_{\rm ref}(z)$ for the resulting unsquared 8×8 fiber. It is Hermitian when $|z| = 1$.

Proposition 2.3 (Reference squared edge)**.** *Let*

$$\eta = 4 + \sqrt{10 + 2\sqrt{5}}.$$

Then

$$\sup_{|z|=1} \rho(A_{\rm ref}(z))^2 = \eta < 8,$$

and equality occurs only at $z = 1$.

Proof. The exact determinant identity is derived and independently certified in Appendix A. We retain here the two algebraic steps that locate the edge. At $z = 1$, put $y = x^2$ and $X = y - 4$. The fiber equation reduces to

$$X^4 - 20X^2 + 80 = 0.$$

Its largest y-root is $4 + \sqrt{10 + 2\sqrt{5}} = \eta$.

For global maximality, set

$$s = \sqrt{10 + 2\sqrt{5}}, \qquad u = y - \eta, \qquad t = 2 - (z + z^{-1}).$$

On the unit circle, $t \in [0, 4]$. The same exact determinant relation, after substitution, is

$$\begin{aligned} 0 = {} & u^4 + 4su^3 + 2u^2 t + (40 + 12\sqrt{5})u^2 + 4sut + 8\sqrt{5}\, su \\ & + t^2 + (4\sqrt{5} - 3)t. \end{aligned}$$

Every displayed coefficient is positive. If a squared fiber eigenvalue satisfied $y \ge \eta$, then $u, t \ge 0$, so the right-hand side could vanish only for $u = t = 0$. Thus $y \le \eta$, and equality forces $z + z^{-1} = 2$, hence $z = 1$. Finally, $\sqrt{5} < 9/4$ and $\sqrt{10 + 2\sqrt{5}} < 191/50$ give $\eta < 391/50 < 8$. $\square$

The reference phase is also the zero-interface member of the gap language. If consecutive positive Q-sites are separated by $g = 4$, the right positive lattice is the continuation of the left lattice in the same translate B_s. Up to translation, reflection, and the lift choice, this is precisely the unperturbed reference bulk, not an abnormal interface.

3 Gaps, Charges, and Translation Sectors

3.1 Cyclic gaps and excess charge

Let $Q \in \{\pm 1\}^n$ be a cyclic quadrilateral-flux word that admits a cyclic τ-lift. Suppose first that its positive set

$$D(Q) = \{i \in \mathbb{Z}/n\mathbb{Z} : Q_i = +1\}$$

is nonempty. Choose integer representatives

$$x_1 < x_2 < \cdots < x_d < x_1 + n$$

for its d elements and put $x_{d+1} = x_1 + n$. The positive cyclic gaps and their excess charges are

$$g_j = x_{j+1} - x_j, \qquad q_j = g_j - 4 \qquad (1 \leq j \leq d).$$

The reference value is four; positive or negative q_j measures the departure of one gap from that spacing.

Proposition 3.1 (Gap and charge identities)**.** *The gap list satisfies*

$$\sum_{j=1}^{d} g_j = n, \qquad \sum_{j=1}^{d} q_j = n - 4d.$$

If n is even and Q has a cyclic τ-lift, then d is even and

$$\sum_{j=1}^{d} q_j \equiv n \pmod 8.$$

Proof. The half-open arcs from x_j to x_{j+1} partition one traversal of the n-cycle, so their lengths sum to n. Subtracting four for each of the d arcs gives the second identity.

The word has d positive and $n - d$ negative entries. Hence

$$\prod_{i=0}^{n-1} Q_i = (-1)^{n-d}.$$

A cyclic lift exists only when this product is 1. Thus $n - d$ is even; since n is even, d is even. Write $d = 2h$. The second identity becomes

$$\sum_{j=1}^{d} q_j = n - 8h,$$

which is the asserted congruence. □

The Hamilton holonomy α does not occur in this argument. It is an independent step-one cycle sign and cannot change the charge law. If $D(Q) = \varnothing$, there is no cyclic gap list and Proposition 3.1 is not invoked; this defect-free word is treated separately whenever it occurs.

3.2 Translation sectors

Recall the four reference translates B_s, $s \in \mathbb{Z}/4\mathbb{Z}$, defined by

$$(B_s)_i = +1 \iff i \equiv s \pmod 4.$$

They are sectors of the Q-word. Each has two period-eight τ-lifts, but the sector label itself depends only on the positions of the positive Q-sites.

Proposition 3.2 (Oriented gap shift)**.** *Suppose an oriented gap of length g leaves the reference sector B_s at a positive site and enters a reference sector on its right. For its charge $q = g - 4$, the right sector is*

$$B_{s+\sigma_{\text{sec}}(q)}, \qquad \sigma_{\text{sec}}(q) = q \pmod 4.$$

This law is independent of the τ-lift and Hamilton holonomy.

Proof. Let the left endpoint be $x \equiv s \pmod 4$. The next positive site is $x + g$, so the positive lattice continuing to the right is

$$x + g + 4\mathbb{Z}.$$

It is therefore B_{s+g}. Since $g = q + 4$, the residues of g and q agree modulo four. The proof uses only endpoint positions, so neither the lift sign nor α enters. □

3.3 Composition and legal cyclic closure

For consecutive oriented gaps with charges $q_1, \ldots, q_k$, repeated use of Proposition 3.2 gives

$$B_s \longmapsto B_{s+q_1} \longmapsto \cdots \longmapsto B_{s+q_1+\cdots+q_k},$$

with labels reduced modulo four. Thus sector shifts add:

$$\sigma_{\text{sec}}(q_1 + \cdots + q_k) \equiv \sum_{j=1}^{k} \sigma_{\text{sec}}(q_j) \pmod 4.$$

On a cyclic gap list, $\sum_j g_j = n$, so the final reference lattice is the translate B_{s+n}; equivalently,

$$\sum_j q_j \equiv n \pmod 4.$$

This compatibility is weaker than the mod-eight law in Proposition 3.1, because it does not contain the parity information needed for a cyclic τ-lift.

The modulo-eight charge closure and the modulo-four translation-sector law encode different constraints and will be used separately. The first is a global condition on a liftable ring; the second records the local reference translate across an interface.

For later use, a gap of length six has charge $q = 6 - 4 = 2$ and sector shift two modulo four. One, two, or three such gaps contribute total charges $2, 4, 6$, respectively, and sector shifts $2, 0, 2$ modulo four. If all other gaps have the reference length four, these are precisely the charge contributions required for ring orders congruent to $2, 4, 6$ modulo eight. No spectral comparison is asserted at this stage.

4 The Elementary Six-Gap Phase Slip

4.1 The bilateral interface and its essential spectrum

4.2 Algebraic candidate and physical matching

4.3 The global G6 edge

4.4 Rank-two symmetry

5 Optimality Among Single-Gap Interfaces

5.1 Canonical single-gap operators

5.2 Exact finite-gap comparisons

5.3 Uniform control of the large gaps

5.4 Reference and single-gap hierarchy

6 Phase Slips on Finite Rings

6.1 Legal residue constructions

6.2 Identification of local patches

6.3 Discrete IMS localization

6.4 Residue upper bounds and the analytic tail

6.5 Structural refinement for separated interfaces

7 Finite Completion of the Classification

7.1 Exact-computation protocol

7.2 Orders 8 through 32

7.3 Orders 34 through 46

7.4 The exact bridge from 48 to 238

7.5 Exhaustive synthesis

8 Concluding Remarks

Data and Code Availability

A Exact Spectral Certification of the Reference Phase and G6

B Exact Finite Classification and Completeness

References

[1] Francesco Belardo, Maurizio Brunetti, and Adriana Ciampella. Unbalanced unicyclic and bicyclic graphs with extremal spectral radius. *Czechoslovak Mathematical Journal*, 71(2):417–433, 2021.

[2] Francesco Belardo, Sebastian M. Cioabă, Jack H. Koolen, and Jianfeng Wang. Open problems in the spectral theory of signed graphs. *The Art of Discrete and Applied Mathematics*, 1(2):P2.10, 2018.

[3] Yonatan Bilu and Nathan Linial. Lifts, discrepancy and nearly optimal spectral gap. *Combinatorica*, 26(5):495–519, 2006.

[4] Maurizio Brunetti and Zoran Stanić. Unbalanced signed graphs with extremal spectral radius or index. *Computational and Applied Mathematics*, 41(3):118, 2022.

[5] Matt DeVos and Robert Šámal. High-girth cubic graphs are homomorphic to the Clebsch graph. *Journal of Graph Theory*, 66(3):241–259, 2011.

[6] Jan Goedgebeur, Jarne Renders, Gábor Wiener, and Carol T. Zamfirescu. K_2-Hamiltonian graphs: II. *Journal of Graph Theory*, 105(4):580–611, 2024.

[7] Jan Goedgebeur and Oliver Schaudt. Exhaustive generation of k-critical H-free graphs. *Journal of Graph Theory*, 87(2):188–207, 2018.

[8] Huiqiu Lin and Bo Ning. A complete solution to the Cvetković–Rowlinson conjecture. *Journal of Graph Theory*, 97(3):441–450, 2021.

[9] Adam W. Marcus, Daniel A. Spielman, and Nikhil Srivastava. Interlacing families I: Bipartite Ramanujan graphs of all degrees. *Annals of Mathematics*, 182(1):307–325, 2015.

[10] Vaibhav Suvagiya. Signed circulants at the Ramanujan bound. arXiv preprint arXiv:2607.18334v1, 2026.